Continually Evolved Feature and Classifiers Learning for Long-Tailed Class-Incremental Remote Sensing Scene Classification

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***Abstract*—Remote sensing data from real-world scenarios manifests a long-tailed distribution, with the continuous emergence of new classes over time. Nevertheless, the existing class-incremental remote sensing classification models neglect the above long-tailed distribution phenomenon, which seriously damages their overall superior performance. Meanwhile, long-tail class-incremental learning (LT-CIL) developed in other areas focuses only on the classifier decision boundary optimization of the tail class, while neglecting the robustness of the feature backbone. The feature backbone trained on the base classes causes a serious significant distribution shift for the incremental classes due to the distributional differences between base and incremental classes. To solve these issues, we propose a continually evolved feature and classifiers learning (CEF-CL) framework for long-tailed class-incremental remote sensing scene classification (LT-CIRSSC). Specifically, tail-class data are scaled and grafted onto head-class data to diversify the semantic information of the tail-class leveraging the rich context of the head classes, which can improve the generalization of the feature backbone. And then, an adaptive multiscale feature fusion (AMFF) module is proposed to couple feature maps of head- and tail-class scale-by-scale for generating virtual tail-class features that deeply perceive head-class information, which can further enhance the reliability of classifier decision boundary optimization. Furthermore, examples from old classes are regarded as pseudo-tail classes to participate in incremental learning, which greatly alleviates catastrophic forgetting of old classes. Extensive experiments on two remote sensing benchmarks demonstrate the superiority of the proposed CEF-CL in comparison with the existing class-incremental learning (CIL).**

***Index Terms*—Catastrophic forgetting, class-incremental learning (CIL), long-tail distribution, remote sensing scene classification (RSSC).**

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I. Introduction

With the rapid advancement of remote sensing interpretation technologies, remote sensing scene classification (RSSC) has become pivotal in applications, such as disaster monitoring, ecological surveillance, natural resource assessment, and urban planning [1]. Recent progress in deep neural networks (DNNs) has spurred the development of numerous DNN-based RSSC models, achieving state-of-the-art performance [2], [3]. For instance, Ji et al. [4] proposed a coarse-to-fine knowledge distillation framework that transfers multigranularity supervisory signals for enhanced feature learning in RSSC. Yang et al. [5] proposed a parameter-efficient adaptation method that enhances large vision models for RSSC through learnable prompt tuning and soft layer adaptation mechanisms. However, in real-world scenarios, new land-cover classes continuously emerge over time. These new classes often exhibit significant distributional differences from existing ones, leading to degraded performance when existing RSSC models are directly applied. Moreover, fine-tuning models with new-class data usually results in catastrophic forgetting of previously learned classes [6]. Consequently, there is a pressing need for RSSC models capable of incrementally learning new classes while retaining knowledge of old ones. Unfortunately, most current RSSC methods are designed for static datasets, limiting their applicability in dynamic real-world environments.

Class-incremental learning (CIL) tackles the challenge of learning from streaming data, enabling DNNs to acquire new classes while preserving knowledge of old ones continuously. Recent studies have extended CIL to remote sensing applications. For instance, Lu et al. [7] introduced a lightweight CIL model for RSSC, leveraging task-specific feature extractors to achieve model efficiency. Liu et al. [8] proposed an open-set incremental learning framework for dynamic remote sensing

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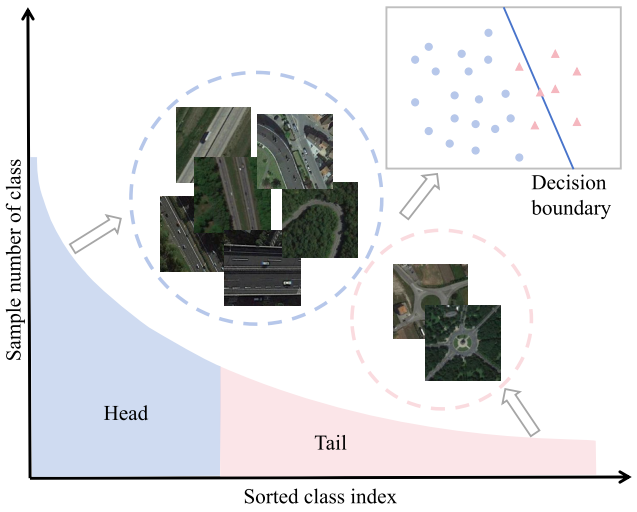


Fig. 1. Long-tail distribution in the remote sensing dataset.

image classification. In addition, Ye et al. [9] enhanced feature extraction in complex environments through efficient channel attention and improved plasticity for new classes via dynamic expansion, while maintaining performance on old classes. However, the above methods assume balanced training data, whereas the real-world remote sensing data often exhibits long-tailed distributions, leading to significant class imbalance [10]. As illustrated in Fig. 1, head classes (e.g., roads, forests, and mountains) are overrepresented due to data availability, while tail classes (e.g., airports) are underrepresented. This imbalance biases models toward head classes, degrading the performance of tail classes and undermining overall effectiveness. Given the prevalence of imbalanced and evolving data in real-world RSSC tasks, developing CIL models robust to long-tailed distributions is critical.

To address the performance degradation of CIL models caused by long-tailed distributions, the concept of long-tail CIL (LT-CIL) was introduced [11]. LT-CIL focuses on enabling DNNs to incrementally learn new classes from long-tailed data, while mitigating catastrophic forgetting of old classes. For instance, Liu et al. [11] employed a decoupled strategy to separately train feature extractors and classifiers, leveraging class-balanced sampling and scaled weight layers to fine-tune classifiers for better adaptation to tail classes. Kalla and Biswas [12] argued that class-balanced sampling compromises head-class performance and instead aligns classifiers using global variance and class prototypes to eliminate adverse effects on head classes. In addition, Chen and Chang [13] proposed a dynamic residual classifier with a lightweight branching layer, enabling the residual fusion of classifier outputs to alleviate data imbalance and address the growing complexity of DNNs. Although they are promising in performance, these methods primarily focus on adjusting decision boundaries for tail-class classifiers, overlooking the robustness of feature backbones. In RSSC tasks, remote sensing data exhibit three distinctive challenges: 1) large intraclass differences from varying resolutions or illumination; 2) high interclass similarity due to overlapping features; and 3) complex background interference with cluttered geospatial patterns. These characteristics intensify in LT-CIL scenarios under data imbalance and distribution shifts, exacerbating feature ambiguity. This inherent ambiguity underscores the need for feature backbones to learn well-structured and coherent representations. Moreover, the significant distributional shift between base and incremental classes leads to severe feature distribution discrepancies when using feature backbones trained on base classes. Therefore, alongside classifier optimization, designing a resilient feature backbone capable of extracting stable and discriminative features is imperative.

To address the challenges in RSSC, we propose a long-tailed class-incremental RSSC (LT-CIRSSC) algorithm based on a continually evolved feature and classifiers learning (CEF-CL) framework. The proposed framework achieves enhanced tail-class generalization through joint optimization of feature backbones and classifiers, effectively addressing the compounded challenges of long-tailed distribution and high interclass similarity inherent in remote sensing imagery. During the feature learning stage, a minority-centric sampling strategy is employed to scale and graft tail-class data into head-class samples, leveraging head-class contextual richness to diversify tail-class semantics. This data hybridization forces discriminative feature learning while alleviating overconfidence in DNNs caused by long-tailed distributions. For the classifier learning stage, an adaptive multiscale feature fusion (AMFF) module is designed to hierarchically couple head–tail feature maps, generating head-aware virtual tail features. These features stabilize at head–tail boundaries, driving optimal boundary adjustments that simultaneously expand tail-class feature space and enhance decision reliability. To bridge the distribution gap between base and incremental classes, a continual dual-learner update is implemented during incremental phases. Old-class exemplars are replayed as pseudo-tail classes, enabling simultaneous acquisition of new knowledge and tail-like old-class retention, thus effectively mitigating catastrophic forgetting. Compared with existing CIL and LT-CIL methods, CEF-CL achieves three critical advancements: 1) mitigating base-incremental class distribution shifts through co-evolving feature-classifier optimization; 2) resolving high interclass similarity and background interference via scaling grafting; and 3) alleviating catastrophic forgetting by treating old-class exemplars as pseudo-tail classes.

The main contributions are presented as follows.

1) We propose a CEF-CL framework for LT-CIRSSC, which preserves old-class performance while adapting to new and long-tailed classes through the progressive evolution of dual learners. To the best of the authors' knowledge, this represents one of the few systematic explorations of LT-CIL paradigms in RSSC.
2) Our dual-adaptation strategy addresses feature backbone robustness and classifier reliability for long-tailed data. A data hybridization technique enriches tail-class semantics by scaling and grafting tail samples into head-class contexts, while an AMFF module generates globally aware virtual tail features through hierarchical head–tail

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feature coupling, enabling more reliable decision boundary optimization.

3) Extensive experiments across multiple scenarios on two public remote sensing datasets demonstrate our framework's superior performance over existing CIL and LT-CIL methods, exhibiting enhanced task generalization capabilities.

II. Related Works

A. Class-Incremental Learning

CIL is a key research paradigm in continual learning [14]. It enables DNNs to progressively learn from sequential tasks with disjoint class sets, while maintaining the accuracy for all encountered classes during inference. The primary challenge in CIL is mitigating catastrophic forgetting of previously learned classes while acquiring new-class knowledge, particularly when the historical training data become inaccessible. Current CIL methods are primarily divided into three categories: regularization-based methods, parameter isolation-based methods, and replay-based methods.

Regularization-based methods primarily constrain parameter updates during new-class learning to preserve previously acquired knowledge. A prominent strategy involves knowledge distillation, applied to both output logits [15] and intermediate layers [16], [17]. Notably, Li and Hoiem [18] pioneered the integration of distillation loss with cross-entropy loss to mitigate forgetting. Hou et al. [19] extended the above approach by incorporating cosine normalization and interclass separation constraints for enhanced old-class preservation. More recently, Choi et al. [20] introduced a dual-teacher distillation framework to simultaneously transfer knowledge from both old and new classes.

Parameter isolation-based methods address catastrophic forgetting by progressively expanding network capacity [21], [22], allocating dedicated parameter spaces for new tasks while preserving learned knowledge. DER [23] introduced a dynamic backbone expansion mechanism with feature aggregation through larger classifiers. Building upon DER, FOSTER [24] incorporated knowledge distillation for model compression to mitigate the network growth. DyTox [25] pioneered the integration of ViT architectures [26] with dynamic expansion strategies for CIL.

Replay-based methods mitigate forgetting by retaining representative samples [19], [27], features [28], or historical models [29]. Zhou et al. [29] established a benchmark for balancing replay examples and model buffers, optimizing storage strategies based on memory constraints. Zhai et al. [30] leveraged masked autoencoders [31] as the foundational framework, enabling the efficient memory utilization through partial image storage while maintaining reconstruction capability.

Replay-based methods have been widely integrated with the aforementioned approaches in CIL. While effective for preserving old classes, these methods typically assume balanced data distributions. They rely on storing a limited number of replay exemplars per old class within a fixed memory budget. A common practice in existing CIL methods is to store $\leq$20 samples per class to mitigate catastrophic forgetting [13], [32], [33]. Coincidentally, classes with $\leq$20 samples are empirically defined as tail classes in long-tailed recognition [34], [35], [36]. Direct utilization of such scarce samples as replay exemplars not only fails to retain old-class knowledge but also degrades incremental learning performance. Our work addresses this by explicitly treating these limited replay exemplars as pseudo-tail classes within our framework. Continuous adaptation of feature extractors and classifiers to tail data enhances the semantic richness of these exemplars during training, effectively preserving old-class performance.

B. Deep Long-Tailed Learning

Traditional deep learning models face significant challenges in handling the prevalent class imbalance in real-world applications. The skewed class distribution biases classifier decision boundaries toward head classes while inadequately representing tail classes, leading to boundary misalignment and performance degradation. Consequently, the deep long-tailed learning has garnered substantial research attention [10], [37]. Current approaches primarily fall into three categories: class rebalancing methods, module improvement methods, and information augmentation methods.

Class rebalancing methods address training sample imbalance through resampling [38], [39] and reweighting techniques [40], [41]. Resampling methods construct balanced training sets via tail-class oversampling or head-class undersampling, while reweighting methods balance loss or gradient contributions by assigning class-specific weights. Module improvement methods employ a decoupled two-stage framework separating feature learning from classifier learning [38], [42], [43], demonstrating superior performance over conventional approaches. This decoupling strategy has been effectively extended to LT-CIL for addressing data imbalance [11], [12], providing the foundational framework for our work.

Recent advances have employed information augmentation methods [44] to enhance the tail-class generalization and overall model performance. For instance, Park et al. [39] implemented CutMix [45] for data augmentation by transferring tail-class foregrounds to head-class backgrounds; however, there is a risk of semantic information loss due to the random region selection. He et al. [46] addressed this limitation through class activation map-guided CutMix [47], preserving semantic integrity at the cost of significant computational overhead. This work introduces ResizeMix [48] during feature learning, which scales and grafts tail-class samples onto head-class backgrounds. This approach maintains semantic information while eliminating additional computational burdens, significantly improving tail-class generalization. Li et al. [34] substituted tail-class features with head-class counterparts during classifier fine-tuning to enrich semantic features, though constrained by single-scale fusion before global pooling. The method establishes multiscale feature coupling between head and tail classes, followed by channel attention-based adaptive fusion of enhanced tail-class features. This multiscale integration better preserves head-class semantics while improving the decision boundary reliability.

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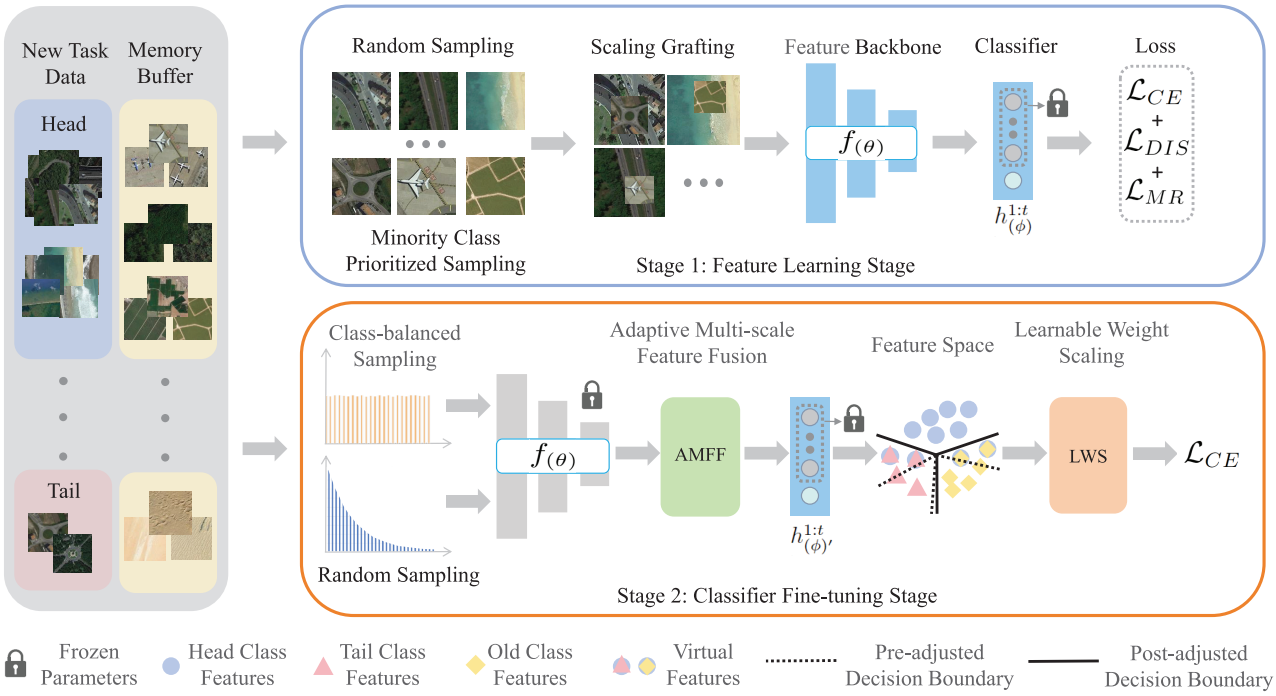


Fig. 2. CEF-CL model framework diagram, AMFF represents the adaptive multiscale feature fusion module and LWS indicates the learnable weight scaling layer. $h^{1:t}_{(\phi)}$ and $h^{1:t}_{(\phi)'}$ represent the classifier initialized and the classifier after fine-tuning, respectively.

III. Methods

A. Problem Definition

In conventional CIL, a DNN model is trained sequentially on disjoint tasks, each introducing a distinct set of classes. Formally, given $\mathcal{T}$ sequential tasks with their data denoted by $\{\mathcal{D}^1, \mathcal{D}^2, \ldots, \mathcal{D}^{\mathcal{T}}\}$, the cumulative class set across all tasks is $\mathcal{C}$. The disjointness property enforces $\mathcal{D}^l \cap \mathcal{D}^k = \emptyset$ for $l \neq k$. During training on task t, the model exclusively utilizes its corresponding dataset $\mathcal{D}^t = \{x_i^t, y_i^t\}_{i=1}^{n^t}$, where x_i^t denotes a remote sensing image, y_i^t denotes its label, and n^t denotes the sample size of task t. In contrast, replay-based CIL methods leverage both current-task data and historical exemplars, i.e., $\mathcal{D}^t \cup \mathcal{M}$, where $\mathcal{M}$ represents a memory buffer storing replay samples (20 per class) from preceding $t-1$ tasks.

In LT-CIL, the training data for each task exhibits class imbalance with exponentially decaying sample sizes across categories [49]. This decay is quantified by the imbalance ratio $\rho = (N_{\min}/N_{\max})$, where $N_{\min}$ and $N_{\max}$ denote the sample counts of the least and most frequent classes, respectively. To address the limitations of existing methods in handling long-tailed incremental tasks, we propose an LT-CIL network model $\Theta = \{f_{(\theta)}, h_{(\phi)}\}$, where $f_{(\theta)}$ is a feature backbone parameterized by θ and $h_{(\phi)}$ represents a classifier with parameters ϕ. Following [11], ordered and shuffled long-tailed scenarios are evaluated without loss of generality. Notably, the testing set remains balanced for fair model evaluation.

B. Framework Overview of CEF-CL

In conventional long-tailed learning, the two-stage decoupled strategy, which separates feature learning from classifier tuning, demonstrates superior performance. Specifically, the process involves two sequential stages:

1) *Feature Learning:* A robust feature backbone is learned via random sampling, with classifier parameters initialized.
2) *Classifier Learning:* The feature backbone is frozen, while the classifier is retrained using class-balanced sampling to achieve balanced decision boundaries.

However, directly applying this strategy to LT-CIL induces catastrophic forgetting, as overemphasizing long-tailed new task data impedes old-class knowledge retention. To address this, the two-stage strategy is enhanced with tail class-specific data mixing and adaptive feature fusion, treating old classes as pseudo-tail classes. This enables continuous joint optimization of both feature backbone $f_{(\theta)}$ and classifier $h_{(\phi)}$ across incremental tasks, simultaneously mitigating catastrophic forgetting and long-tailed distribution challenges.

As illustrated in Fig. 2, during task t, model Θ accesses both new task data $\mathcal{D}^t$ and old-class replay exemplars $\mathcal{M}$. The key innovations are as follows.

1) *Feature Learning Stage:* A scaling grafting method replaces traditional data augmentation to enhance the tail-class generalization, coupled with data mixing for robust feature distribution learning.
2) *Classifier Fine-Tuning Stage:* Multiscale feature fusion adaptively strengthens tail-class features, refining decision boundaries.

The following will detail the training workflow of the proposed CEF-CL framework on incremental task t.

C. Feature Learning Stage

To address the limited tail-class adaptability of the feature backbone and tail-class misclassification due to high inter-class similarity in the remote sensing data under conventional

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long-tailed learning, a scaling grafting method is proposed during the feature learning stage. Following recent long-tailed learning research [34], [35], [36], we define classes with over 100 samples as head classes, classes with not more than 20 samples as tail classes, and classes with sample counts between 20 and 100 as medium classes. Specifically, scaled data biased toward minority classes (tail and medium classes) are sampled via inverse class-frequency weighting, while grafted target data biased for head classes are selected through random sampling. For each class $c \in (1, \mathcal{C})$, the minority-prioritized sampling probability p_t^c and random sampling probability p_i^c are computed as follows:

$$p_t^c = \frac{1/n^c}{\sum_{c'=1}^{\mathcal{C}} 1/n^{c'}}, \quad p_i^c = \frac{n^c}{N} \tag{1}$$

where n^c and $n^{c'}$ denote the sample counts of classes c and c', respectively, and N represents the total number of samples in the training set. Using the two sampling strategies, we obtain minority class-centered data $\mathcal{S}_T$ and head class-centered data $\mathcal{S}_I$ from the dataset. Taking $(x_t, y_t) \sim \mathcal{S}_T$ and $(x_i, y_i) \sim \mathcal{S}_I$ as examples, the scaling grafting method is detailed as follows: x_t represents the scaled image sample, x_i denotes the target image sample for grafting, and their combination yields a new sample $(\hat{x}, \hat{y})$, formulated as follows:

$$\begin{aligned} \hat{x} &= \text{Paste}\left(x_t \cdot \gamma, x_i, R_i\right) \\ \hat{y} &= \lambda y_t + (1 - \lambda)\, y_i \end{aligned} \tag{2}$$

where $\hat{x}$ represents the new sample image generated by mixing the scaled sample image x_t and the target sample image x_i, with the corresponding soft label $\hat{y}$ determined by the mixing ratio of the original labels. The scaling rate γ is sampled from a uniform distribution $\gamma \sim U(\alpha, \beta)$, where α and β denote the lower and upper bounds of the scaling range, respectively. These bounds ensure that tail-class samples are not scaled to structurally invalid sizes, preserving discriminative visual patterns during augmentation. R_i indicates the pasting region in the target image x_i, whose size matches the scaled x_t. The image mixing ratio λ, defined as the size ratio of the scaled image to the target image, ranges in (0, 1) and is computed as follows:

$$\lambda = \frac{(W_t \cdot \gamma) \cdot (H_t \cdot \gamma)}{W_i \cdot H_i} \tag{3}$$

where W_t and H_t denote the width and height of the scaled sample image, respectively, W_i and H_i represent the width and height of the target image to be grafted, respectively, and γ indicates the scaling factor. It is crucial to emphasize that our method preserves the complete semantic integrity of the tail-class sample x_t through holistic scaling, eliminating CutMix's risk of cropping discriminative features from scarce tail-class instances. By intentionally grafting the scaled tail-class sample onto randomly located R_i regions of head-class images, we leverage the inherent feature abundance and robustness of head classes to enrich tail-class representations within diverse contextual environments.

Compared with conventional augmentation methods (e.g., Mixup and CutMix), scaling grafting better addresses the unique challenges of remote sensing images. Mixup often generates unnatural or uninterpretable blended images, while CutMix tends to crop background regions, causing semantic inconsistency between mixed images and their labels. Remote sensing images, characterized by low spatial resolution and diverse ground object types with blurred boundaries, suffer severe background interference, issues exacerbated by directly applying Mixup/CutMix. As shown in Fig. 3, Mixup produces blurred images, whereas CutMix erroneously combines airport and forest labels when pasting highway backgrounds. The method adaptively scales and grafts images to preserve core regions, effectively avoiding unnatural blending and background interference. By proportionally scaling minority-class samples and grafting them onto head-class targets, tail-class diversity is enhanced through head-class contextual information. This enables the feature backbone $f_{(\theta)}$ to learn robust tail-class features and discriminative head–tail features, simultaneously improving generalization on tail classes and handling high interclass similarity in remote sensing data.

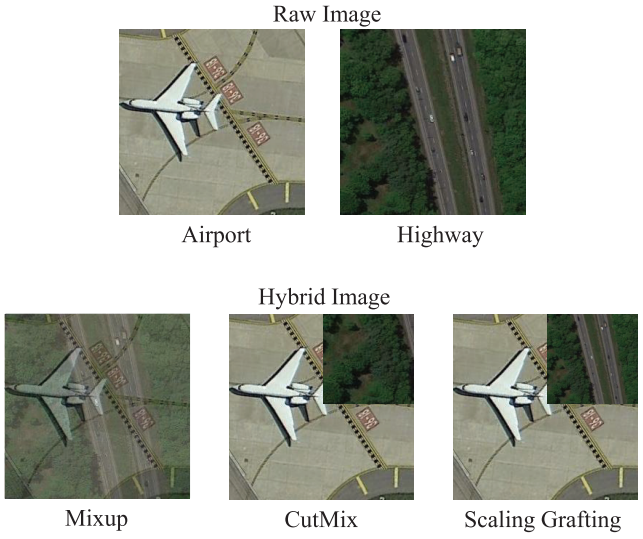


Fig. 3. Comparison of different data augmentation.

For classifier design, we deploy task-specific classifiers $h_{(\phi)}^t$ for each incremental stage, whose output dimension matches the class count of task t. During task t, only parameters of $h_{(\phi)}^t$ are updated while freezing those of prior classifiers $h_{(\phi)}^{1:t-1}$ to preserve old-class knowledge. The cross-entropy loss for the initial task t is defined as follows:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{D}^t| + |\mathcal{M}|} \sum_{(x,y) \in \mathcal{D}^t \cup \mathcal{M}} y \cdot \log\left(p_x\right) \tag{4}$$

where y is a one-hot vector and p_x denotes the probability vector over all classes before task t. Class imbalance biases old-class classifier weights and feature representations, causing misclassification of old classes as new ones [19]. To mitigate this, distillation loss $\mathcal{L}_{\text{DIS}}$ and interclass margin loss $\mathcal{L}_{\text{MR}}$ are introduced during feature learning, as formulated in (5) and (6), to preserve discriminative knowledge between old and new classes

$$\mathcal{L}_{\text{DIS}} = 1 - \left\langle f_{(\theta)}^*(x), f_{(\theta)}(x) \right\rangle \tag{5}$$

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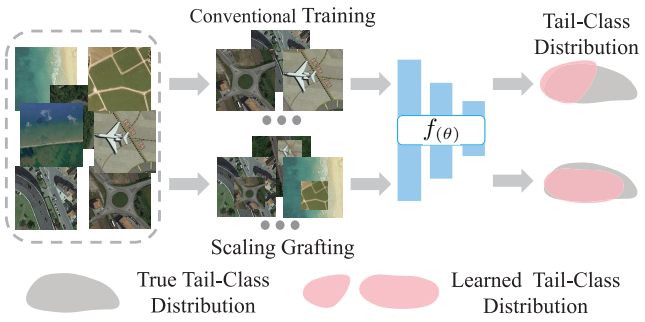


Fig. 4. Comparison of tail-class distribution.

where $f^*_{(\theta)}(x)$ and $f_{(\theta)}(x)$ represent the features of sample x extracted by the old and new models, respectively. The distillation loss $\mathcal{L}_{\text{DIS}}$ preserves old-class knowledge by minimizing the cosine distance between the features of the new and old models

$$\mathcal{L}_{\text{MR}} = \sum_{k=1}^{K} \max\left(m - \langle h^x_{(\phi)}, f_{(\theta)}(x)\rangle + \langle h^k_{(\phi)}, f_{(\theta)}(x)\rangle, 0\right). \tag{6}$$

The margin loss $\mathcal{L}_{\text{MR}}$ enforces a separation between new-class and old-class classifier weights, preventing the misclassification of old classes as new ones. Here, m is the margin threshold, $h^x_{(\phi)}$ denotes the classifier weight for the ground-truth class of x, and $h^k_{(\phi)}$ represents the classifier weight of the top-kth hard negative class.

Notably, the scaling grafted sample label $\hat{y}$ is formulated as a soft label, linearly interpolating the original labels based on the image mixing ratio λ. Specifically, the loss computation incorporates this blending factor as follows:

$$\mathcal{L}_{\hat{y}} = \lambda \cdot \mathcal{L}(z, y_t) + (1-\lambda) \cdot \mathcal{L}(z, y_i) \tag{7}$$

where z is the output of the feature learning model Θ for the mixed sample $\hat{x}$. These scaling grafted samples not only enhance the model's ability to fit tail classes toward their original distribution (see Fig. 4) but also alleviate overconfidence bias induced by long-tailed data.

D. Classifier Fine-Tuning Stage

To alleviate decision boundary shifts caused by long-tailed data in incremental tasks, an AMFF module is proposed to enrich tail-class features, effectively improving the performance of the classifier. Inspired by [34], head–tail feature coupling is introduced during classifier fine-tuning, leveraging head-class semantics to diversify tail-class features and recalibrate decision boundaries. Specifically, primary branch: class-balanced sampled data $\mathcal{S}_B$ and secondary branch: randomly sampled data $\mathcal{S}_I$ (head-class biased). For class $c \in \mathcal{C}$, the class-balanced sampling probability p^c_b is computed as follows:

$$p^c_b = \frac{1}{\mathcal{C}}. \tag{8}$$

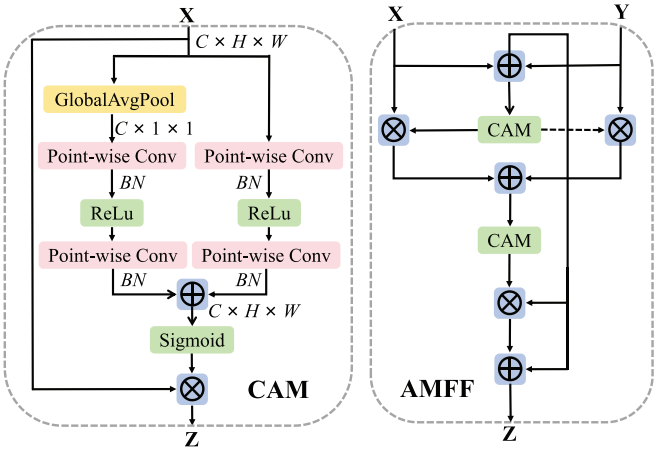


Fig. 5. AMFF module framework diagram.

The class-balanced data $\mathcal{S}_B$ and randomly sampled data $\mathcal{S}_I$ are processed by a frozen feature backbone $f_{(\theta)}$ to obtain features F_B and F_I, respectively. The fusion is as follows:

$$\hat{F} = M_\tau \odot F_B + (1 - M_\tau) \odot F_I \tag{9}$$

where $M_\tau \in \{0,1\}^{W_F \times H_F \times C}$ is a channelwise binary mask generated by randomly selecting $\tau \times C$ channels to set to 1 (with all spatial positions in these channels set to 1) and setting the remaining channels to 0. Here, $\tau \in [0,1]$ controls the fusion ratio by determining the proportion of channels taken from F_B, C is the number of feature map channels, and $\odot$ denotes elementwise multiplication. The complement mask $1 - M_\tau$ ensures disjoint coverage by activating the complementary set of channels.

Existing methods suffer from unreliable decision boundary adjustment due to single-scale head–tail feature fusion. The AMFF module addresses the matter via multiscale coupling, aggregating F_B and F_I across multiple scales to generate semantics-rich virtual features, and iterative attention, adaptively fusing multiscale features through attention mechanisms, enhancing decision boundary reliability. Specifically, AMFF first couples head-class semantics from $\mathcal{S}_I$ with tail-class features from $\mathcal{S}_B$ at multiple scales, which generates virtual features for preserving hierarchical semantic relationships. Subsequently, an attention-guided fusion iteratively weights scale-specific features based on their discriminative power.

The AMFF module adaptively fuses multiscale virtual features through iterative channel attention mechanisms, as illustrated in Fig. 5. To mitigate the substantial influence of initial fusion quality on final weight determination, fusion features are first initialized using a channel attention module (CAM). Subsequent attention processing yields stabilized channel weights, and skip connections preserve original fusion features, which enhances the hierarchical information utilization and prevents over-adjustment. CAM employs dual-scale branches to extract channel attention weights: global average pooling (GlobalAvgPool) aggregates spatial information, while 1×1 convolutions (pointwise Conv) model channelwise relationships, followed by batch normalization (BN), ReLU activation, and Sigmoid normalization to constrain weights

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within (0, 1). The final output Z is obtained by elementwise multiplication between the input X and learned weights. In ResNet implementation, head-class and tail-class features from the third and fourth residual blocks are fused via (9), which generates multiscale virtual tail features $\hat{F}_x$ and $\hat{F}_y$. This is because the third and fourth residual blocks in ResNet provide complementary feature representations. The fourth residual block extracts high-level semantic features due to its larger receptive fields, while the third residual block retains richer spatial details critical for distinguishing subtle interclass variations. Integrating these two levels enables the model to leverage broader and richer feature representations. Then, these two features are processed through the AFFM module to derive the ultimate virtual tail feature $\hat{F}_z$, formally expressed as follows:

$$\begin{aligned}\hat{F}_m &= \hat{F}_x + \hat{F}_y \\ \hat{F}_{xy} &= \mathrm{CA}\left(\hat{F}_m\right) \odot \hat{F}_x + \left(1 - \mathrm{CA}\left(\hat{F}_m\right)\right) \odot \hat{F}_y \\ \hat{F}_z &= \hat{F}_m + \mathrm{CA}\left(\hat{F}_{xy}\right) \odot \hat{F}_m \end{aligned} \tag{10}$$

where $\mathrm{CA}(\hat{F}_m)$ denotes the channel attention weights (ranging $[0, 1]$) for feature $\hat{F}_m$, with $1 - \mathrm{CA}(\hat{F}_m)$ being their complements as shown by dashed arrows in Fig. 5. The elementwise multiplication $\odot$ generates virtual tail-class features $\hat{F}_z$. The CAM adaptively balances channel contributions via dual-scale branches, which analyze semantic (fourth block) and spatial (third block) feature importance. This selective fusion generates robust fused features that integrate both hierarchical semantics and localized details. These features anchor near decision boundaries, shifting head-class boundaries toward regions favorable to tail/old classes, thereby enhancing the reliability of boundary calibration.

The classifier $h^t_{(\phi)}$ updates its weights based on these virtual features, yielding $h^t_{(\phi)'}$. To mitigate long-tailed data bias, a learnable weight scaling (LWS) layer is introduced, with weight parameters $\mathcal{W}$ being the number of classes before task t. The final output $\hat{z}$ of the CEF-CL model is computed as follows:

$$\hat{z} = \mathcal{W} \odot h^{1:t}_{(\phi)'}\left(f^t_{(\theta)}(x)\right) \tag{11}$$

where $\odot$ denotes the elementwise multiplication and the cross-entropy loss is exclusively employed during classifier fine-tuning.

IV. Experiment

A. Datasets

The method is evaluated on two public remote sensing datasets: NWPU-RESISC45 [50] and AID [52]. NWPU-RESISC45 contains 45 classes with 700 samples per class, split into training and testing sets at an 8:2 ratio. AID comprises 30 classes, with 220–420 samples per class; 100 samples per class are held out for testing, and the remainder are used for training. Both datasets are converted to long-tailed versions using a long-tailed data loader with an imbalance ratio $\rho = 0.01$, while the testing sets remain balanced.

B. Experimental Setting

Following [11], approximately half of the dataset classes are allocated to the base task, with the remainder evenly split across incremental tasks. Specifically, for NWPU-RESISC45, 25 classes form the base task, and the remaining 20 classes are divided into five or ten incremental tasks. For AID, 15 classes form the base task, and the remaining 15 classes are grouped into three or five incremental tasks.

During the feature learning stage, the initial learning rate is set to 0.5, iterations to 100, scaling bounds $\alpha = 0.1$ and $\beta = 0.8$, and the margin threshold m in $\mathcal{L}_{\mathrm{MR}}$ to 0.5. For the classifier fine-tuning stage, the initial learning rate is 0.5, iterations are 30, and the fusion ratio $\tau = 0.3$; both stages use a batch size of 32. For a fair comparison, all methods adopt ResNet34 [53] as the backbone, with 20 replay samples per class selected via the herding algorithm. Evaluation is done using average incremental accuracy [54], defined as the mean accuracy across all incremental tasks.

For baselines, DER, FOSTER, and MEMO are implemented using PyCIL [55]; UCIR+LT-CIL, MAFDRC, DGR, and APART use their original code; and other methods are based on the implementation in [11].

C. Comparison With State-of-the-Art CIL Methods

To validate the effectiveness of the CEF-CL framework, we compare it with state-of-the-art CIL and LT-CIL methods on NWPU-RESISC45 and AID datasets under multiple scenarios (see Table I). The CEF-CL consistently outperforms all baselines in both ordered and shuffled LT-CIL. Specifically, on the NWPU-RESISC45 dataset with five incremental tasks, CEF-CL achieves improvements of +1.39% in shuffled LT-CIL and +1.58% in ordered LT-CIL compared with the best baseline. Similarly, on the AID dataset with three incremental tasks, CEF-CL surpasses the best baseline by +1.19% and +1.75% in shuffled and ordered LT-CIL, respectively. Analysis reveals that CEF-CL exhibits stronger robustness compared with existing methods. For instance, DER [23] demonstrates suboptimal performance in ordered LT-CIL but suffers severe degradation in shuffled LT-CIL. Similarly, MAFDRC [13] performs well on NWPU-RESISC45 but struggles on AID. These limitations stem from two key issues: 1) traditional CIL methods fail to account for long-tailed data distributions, leading to poor adaptability in severe imbalance scenarios and 2) existing LT-CIL methods overly focus on classifier tuning and neglect the robustness of the feature backbone, resulting in limited generalization across datasets. CEF-CL addresses these challenges through a dual strategy: 1) tail-class-specific data blending and multiscale feature fusion to enhance feature robustness and 2) joint optimization of the feature extractor $f_{(\theta)}$ and classifier $h_{(\phi)}$ while preserving old-class knowledge. The approach ensures the superior performance in LT-CIRSSC tasks across diverse datasets and scenarios.

D. Ablation Studies

1) Contribution of Different Components: To evaluate the impact of the two-stage integration of scaling grafting and AMFF module in the CEF-CL framework, ablation studies

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TABLE I

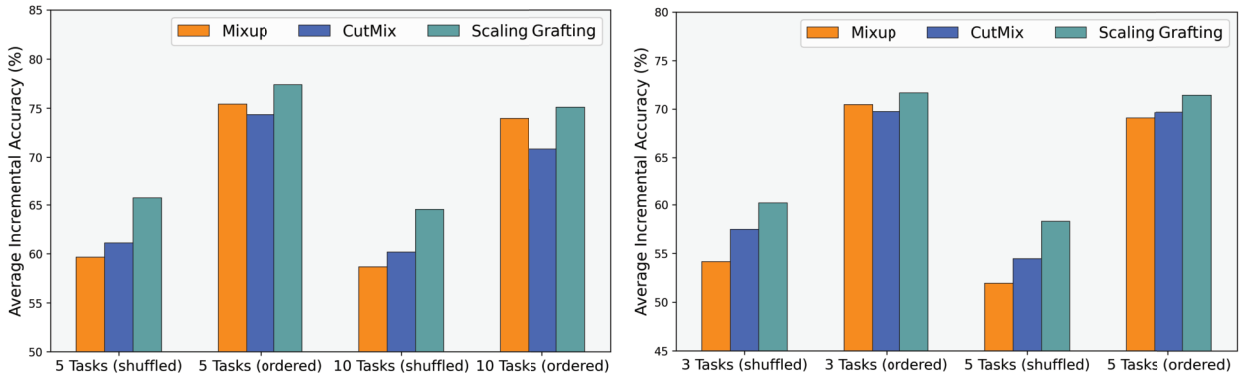
EXPERIMENTAL PERFORMANCE COMPARISON WITH DIFFERENT ALGORITHMS ON SHUFFLED LT-CIL AND ORDERED LT-CIL

Datasets	NWPU-RESISC45				AID			
Experimental Setting	5 Incremental Tasks		10 Incremental Tasks		3 Incremental Tasks		5 Incremental Tasks	
	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered
LWF (TPAMI 2018) [18]	54.69	66.86	53.71	63.20	38.96	55.15	56.13	66.55
EEIL (ECCV 2018) [15]	55.10	67.44	53.00	67.11	39.08	51.68	56.63	68.14
LUCIR (CVPR 2019) [19]	55.31	69.66	52.43	68.92	52.40	63.90	56.88	69.13
DER (CVPR 2021) [23]	59.81	75.80	57.34	74.27	41.94	61.99	40.95	61.00
SSIL (ICCV 2021) [27]	45.46	59.96	45.35	58.48	43.73	55.43	41.27	54.54
FOSTER (ECCV 2022) [24]	60.32	73.01	54.47	74.61	54.67	69.74	49.94	69.78
EEIL+LT-CIL (ECCV 2022) [11]	53.68	60.05	52.73	59.76	43.45	51.33	45.65	52.52
LUCIR+LT-CIL (ECCV 2022) [11]	52.02	71.15	50.90	68.98	55.25	67.12	54.51	67.06
MAFDRC (ICCV 2023) [13]	61.42	72.63	60.76	65.87	44.55	55.60	41.89	55.51
MEMO (ICLR 2023) [29]	60.51	70.71	61.13	71.96	48.89	69.93	50.67	70.90
UCIR+GVAlign (WACV 2024) [12]	55.22	67.62	54.00	66.15	58.97	67.98	57.47	67.20
DGR (CVPR 2024) [50]	55.08	56.08	51.35	46.08	58.17	58.93	57.62	57.01
APART (ML 2025) [51]	64.33	73.10	59.82	64.33	59.01	67.21	57.39	63.33
CEF-CL (ours)	**65.72**	**77.38**	**64.53**	**75.02**	**60.20**	**71.68**	**58.30**	**71.40**

TABLE II

ABLATION EXPERIMENT OF THE PROPOSED CEF-CL FRAMEWORK ON SHUFFLED LT-CIL AND ORDERED LT-CIL

Datasets	NWPU-RESISC45				AID			
Experimental Setting	5 Incremental Tasks		10 Incremental Tasks		3 Incremental Tasks		5 Incremental Tasks	
	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered
CEF-CL (w/o Scaling Grafting)	53.57	73.23	51.80	71.98	54.05	66.73	53.81	67.32
CEF-CL (w/o AMFF)	62.38	75.21	63.57	72.56	56.55	68.40	55.53	68.96
CEF-CL	**65.72**	**77.38**	**64.53**	**75.02**	**60.20**	**71.68**	**58.30**	**71.40**



(a) (b)

Fig. 6. Experimental performance comparison of Mixup, CutMix, and scaling grafting on shuffled LT-CIL and ordered LT-CIL. (a) NWPU-RESISC45 dataset. (b) AID dataset.

were performed on the NWPU-RESISC45 and AID datasets under multiple scenarios (see Table II). The CEF-CL (w/o scaling grafting) removes the scaling grafting operation during feature learning, relying solely on random sampling for training. At the same time, CEF-CL (w/o AMFF) excludes the AMFF module during classifier fine-tuning, using only class-balanced sampling-derived features for boundary adjustment. Experimental results demonstrate that omitting scaling grafting causes significant performance degradation across all scenarios, highlighting its critical role in enhancing the feature backbone's discriminative capability and tail-class generalization. Similarly, removing the AMFF module leads to a consistent accuracy drop of 2%~3% in both datasets, underscoring its importance in stabilizing decision boundary refinement. These findings validate the necessity of both components for robust feature learning and reliable classifier adaptation in LT-CIRSSC tasks.

2) Analysis on Different Mixing Methods: To verify the enhanced applicability of scaling grafting in long-tailed remote sensing applications, we compare it with conventional augmentation methods (Mixup and CutMix) by replacing the scaling grafting component in CEF-CL. As shown in Fig. 6, experimental results on NWPU-RESISC45 and AID datasets show that scaling grafting outperforms both benchmarks,

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TABLE III
EXPERIMENTAL PERFORMANCE COMPARISON WITH FOUR DIFFERENT IMBALANCE RATIOS IN THE NWPU-RESISC45 DATASET

Imbalance Ratios	ρ=0.01		ρ=0.05		ρ=0.1		ρ=0.5	
Experimental Setting	5 Incremental Tasks							
	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered
LUCIR+LT-CIL (ECCV 2022) [11]	52.02	71.15	65.58	78.42	71.60	80.18	77.60	78.35
UCIR+GVAlign (WACV 2024) [12]	55.22	67.62	69.51	78.38	74.80	80.65	80.52	80.25
CEF-CL (ours)	**65.72**	**77.38**	**78.78**	**83.28**	**83.60**	**84.62**	**86.58**	**86.02**

TABLE IV
EXPERIMENTAL PERFORMANCE COMPARISON WITH FOUR DIFFERENT INCREMENTAL TASKS IN THE NWPU-RESISC45 DATASET

Experimental Setting	2 Incremental Tasks		5 Incremental Tasks		7 Incremental Tasks		10 Incremental Tasks	
	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered
LUCIR+LT-CIL (ECCV 2022) [11]	50.13	71.57	52.02	71.15	52.18	68.80	50.90	68.98
UCIR+GVAlign (WACV 2024) [12]	57.13	69.33	55.22	67.62	55.39	67.02	54.00	66.15
CEF-CL (ours)	**63.80**	**77.00**	**65.72**	**77.38**	**65.15**	**74.76**	**65.43**	**75.02**

achieving significant performance gains (2.73%–4.61% higher accuracy) in shuffled long-tailed scenarios with severe class imbalance. Notably, Mixup demonstrates competitive accuracy in ordered long-tailed scenarios but suffers substantial degradation in shuffled long-tailed scenarios. The effectiveness of scaling grafting originates from its unique mechanism: directly integrating rescaled tail-class samples into head-class data. This approach simultaneously reduces the background interference in remote sensing images and improves the tail-class generalization. These advantages enable consistent performance superiority across all testing scenarios, confirming scaling grafting as the optimal mixing method for long-tailed remote sensing classification compared with Mixup and CutMix.

E. Parameters Sensitivity

To investigate the impact of various hyperparameters on model accuracy, extensive experiments were conducted on the NWPU-RESISC45 dataset, varying imbalance ratios, the number of incremental tasks, base classes, resizing scales, and fusion ratios. In addition, we compared our approach with two recent LT-CIL methods, LUCIR+LT-CIL and UCIR+GVAlign, to assess their relative performance.

1) Performance Comparison Under Different Imbalance Ratios: To assess the impact of long-tailed data on the CEF-CL framework, its performance was evaluated on the NWPU-RESISC45 dataset across five incremental tasks under four imbalance ratios ρ, where a higher ρ indicates a more balanced training distribution. Table III presents the comparison between CEF-CL and two state-of-the-art methods, LUCIR+LT-CIL [11] and UCIR+GVAlign [12]. The results demonstrate that CEF-CL consistently outperforms both baselines across all imbalance settings in both shuffled and ordered LT-CIL, with its advantage becoming more pronounced as ρ decreases. Specifically, when ρ = 0.01, CEF-CL achieves classification accuracy gains of 13.7% and 10.5% over LUCIR+LT-CIL and UCIR+GVAlign, respectively, in the shuffled setting, and 6.23% and 9.76% in the ordered setting. This superiority stems from the fact that existing methods focus primarily on classifier adjustment while overlooking the generalization ability of the feature backbone, leading to tail-class misclassification in remote sensing data with high interclass similarity. In contrast, CEF-CL leverages a data-mixing strategy during feature learning, enabling the feature backbone to capture richer tail-class semantics and better discriminate interclass variations, thereby enhancing its adaptability to long-tailed remote sensing classification.

2) Performance Comparison Across Different Incremental Tasks: To evaluate the robustness of CEF-CL under varying incremental settings, we tested its performance on the NWPU-RESISC45 dataset with 2, 5, 7, and 10 incremental tasks. As described in the experimental setting, 25 classes were selected as base classes, while the remaining 20 classes were evenly distributed across incremental tasks. Specifically, for the seven-task setting, the first six incremental tasks contained three classes each, and the last task contained two. As shown in Table IV, as the number of incremental tasks increases, LUCIR+LT-CIL [11] and UCIR+GVAlign [12] suffer from catastrophic forgetting due to parameter updates, leading to a progressive decline in classification performance. In contrast, CEF-CL maintains stable accuracy across all incremental settings, outperforming the baselines by approximately 10% in the shuffled LT-CIL and 7% in the ordered LT-CIL. This highlights the effectiveness of our two-stage strategy, which integrates data mixing and feature fusion for tail classes, mitigating catastrophic forgetting and preserving old-class knowledge.

3) Performance Comparison Across Different Base Class Numbers: To evaluate the impact of base class selection on incremental learning, we conducted experiments on the NWPU-RESISC45 dataset with 10, 15, 20, and 25 base classes, allocating the remaining classes to five incremental tasks. As shown in Table V, while all methods benefit from an increasing number of base classes, especially in the

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TABLE V
EXPERIMENTAL PERFORMANCE COMPARISON WITH FOUR DIFFERENT BASE CLASS NUMBERS IN THE NWPU-RESISC45 DATASET

Number of base classes	10 classes		15 classes		20 classes		25 classes	
Experimental Setting	5 Incremental Tasks							
	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered	Shuffled	Ordered
LUCIR+LT-CIL (ECCV 2022) [11]	43.80	64.45	42.87	68.57	48.20	70.30	52.02	71.15
UCIR+GVAlign (WACV 2024) [12]	50.47	70.75	48.68	71.90	52.98	69.77	55.22	67.62
CEF-CL (ours)	**57.60**	**76.97**	**56.00**	**77.82**	**64.65**	**77.05**	**65.72**	**77.38**

shuffled LT-CIL, CEF-CL consistently outperforms competing approaches across all settings. Specifically, compared with the second-best method, UCIR+GVAlign [12], CEF-CL achieves accuracy gains of 7.13%, 7.32%, 11.67%, and 10.50% for 10, 15, 20, and 25 base classes, respectively, in the shuffled scenario, and 6.02%, 5.92%, 7.28%, and 9.76% in the ordered scenario. Notably, the performance gap widens significantly when handling complex long-tailed distributions with 20+ base classes, demonstrating CEF-CL's enhanced capacity to leverage hierarchical semantic relationships through the multiscale feature fusion. These results validate the effectiveness of our scaling grafting strategy, which enables the feature backbone to learn robust features even with fewer base classes while continuously enhancing its adaptability to long-tailed distributions in subsequent incremental tasks. Furthermore, with a larger number of base classes, CEF-CL fully leverages the expanded feature space to train a more generalizable backbone, further improving performance.

4) Impact of Resizing Scale Range: This section investigates the impact of resizing scale ranges in scaling grafting. Given that the scaling rate γ is sampled from a uniform distribution $\gamma \sim U(\alpha,\beta)$, systematic evaluations of α and β combinations are conducted to analyze γ range constraints. Theoretical analysis reveals that overly small α risks losing tail-class discriminative details. At the same time, substantial β causes tail-class samples to occlude head-class regions. Experimental results for 16 distinct resizing scale ranges $\alpha \in [0.05, 0.3]$ and $\beta \in [0.6, 0.9]$ under shuffled LT-CIL with five incremental tasks on the NWPU-RESISC45 dataset are presented in Fig. 7. Experiments confirm $\alpha = 0.1$ and $\beta = 0.8$ deliver optimal accuracy by balancing these effects. These empirically determined parameters are subsequently adopted as default settings for all scaling grafting implementations.

5) Impact of Fusion Ratio: To investigate the impact of fusion ratio τ on experimental results, comprehensive experiments are conducted across all scenarios of the NWPU-RESISC45 dataset. Fig. 8 illustrates the variation of average incremental accuracy with different fusion ratios, revealing a sharp performance decline across all scenarios when $\tau > 0.3$. This deterioration stems from excessive fusion of head-class features into virtual tail-class representations, which severely erodes the feature space of head classes and consequently degrades their classification accuracy. The optimal performance across various long-tailed scenarios is achieved at $\tau = 0.3$. The results highlight the critical role of fusion ratio selection in preserving head-class samples while effectively capturing tail-class patterns. The systematic analysis of fusion ratios' correlation with feature distribution facilitates model optimization and enhances generalization across application scenarios.

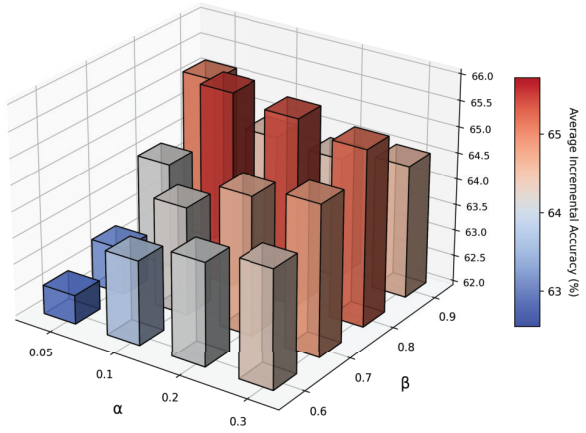


Fig. 7. Average incremental accuracy on the NWPU-RESISC45 dataset versus resizing scale range α and β on shuffled LT-CIL.

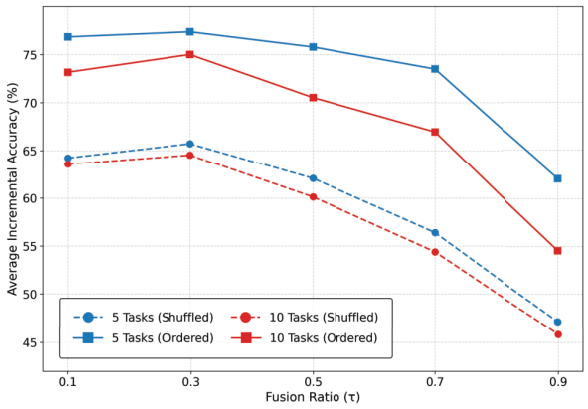
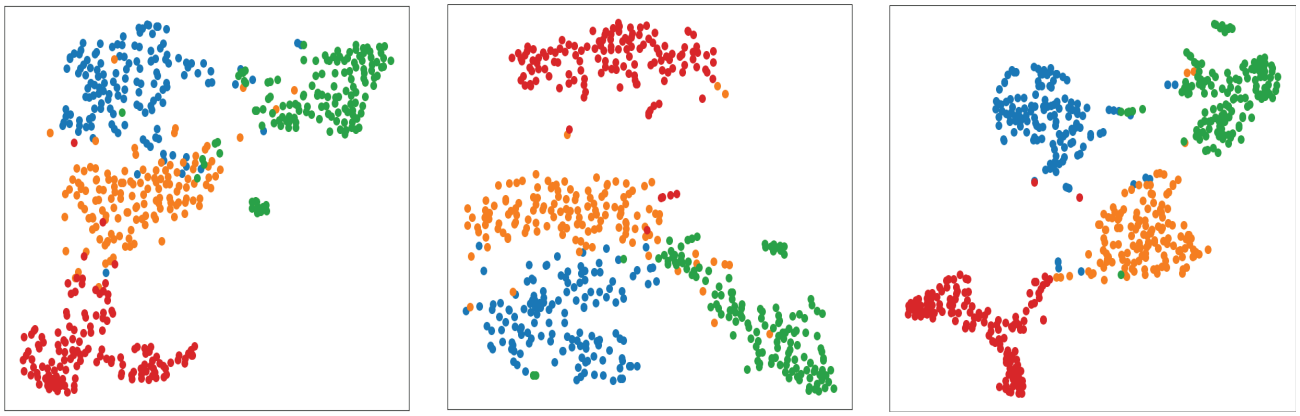


Fig. 8. Experimental performance comparison with five different fusion ratios in the NWPU-RESISC45 dataset.

F. Analysis of Computational Complexity

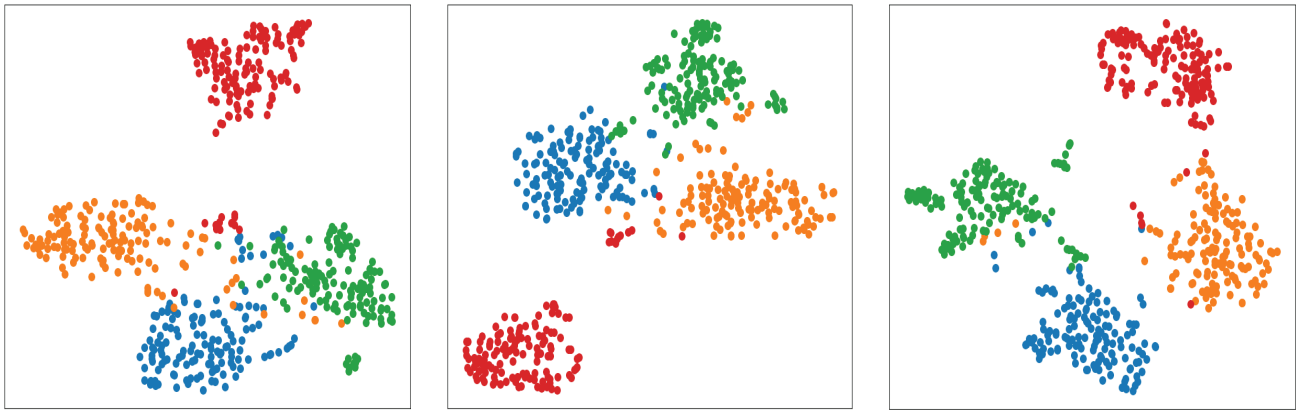
To further validate the computational efficiency of the AMFF module, we conduct a dedicated theoretical and empirical analysis of its time complexity. The dual-branch architecture from CAM introduces a controlled computational overhead characterized by the FLOPs formulation $(4C^2/r)(HW+1)$, where C, H, and W denote the input feature

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(a) (b) (c)

Fig. 9. t-SNE visualization performance comparison of the third shuffled LT-CIL session in the NWPU-RESISC45 dataset. (a) LUCIR+LT-CIL [11]. (b) UCIR+GVAlign [12]. (c) CEF-CL (Ours).



(a) (b) (c)

Fig. 10. t-SNE visualization performance comparison of the third-ordered LT-CIL session in the NWPU-RESISC45 dataset. (a) LUCIR+LT-CIL [11]. (b) UCIR+GVAlign [12]. (c) CEF-CL (Ours).

TABLE VI

COMPLEXITY ANALYSIS OF CEF-CL ON THE NWPU-RESISC45 DATASET WITH FIVE INCREMENTAL SHUFFLED LT-CIL TAKS

Method	Params (M)	Time (h)
CEF-CL (w/o AMFF)	21.3	1.85
CEF-CL	21.8	2.03

map's channel count, height, and width, respectively, and r controls channel dimension reduction via fully connected layers to mitigate computational costs. As shown in Table VI, experimental measurements under the shuffled LT-CIL (five tasks on NWPU-RESISC45) demonstrate that the AMFF module increases model parameters by only 0.53M (2.5% relative growth) while extending training time by 9.7%. This modest overhead is operationally justified by a 3.34% accuracy improvement in this challenging setting, confirming that the AMFF module achieves favorable accuracy-complexity trade-offs without compromising training scalability.

G. t-SNE Visualization

To provide a more intuitive analysis of the CEF-CL framework, t-SNE is employed to visualize feature embeddings under five incremental tasks of the NWPU-RESISC45 dataset. Specifically, we analyze the feature distributions in the third incremental task under both shuffled LT-CIL and ordered LT-CIL scenarios, using LUCIR+LT-CIL [11] and UCIR+GVAlign [12] as comparison methods. The visualization results are presented in Figs. 9 and 10, where different colors represent different classes. As shown in the figures, in the shuffled LT-CIL, CEF-CL exhibits superior clustering performance compared with the baselines. In the ordered LT-CIL, all three methods achieve reasonable clustering, but CEF-CL provides better class separability with more distinct decision boundaries. These results demonstrate that CEF-CL adapts more effectively to long-tailed data. Furthermore, the feature fusion operation introduced during classifier fine-tuning further expands the feature space of tail classes, improving class separability and enhancing decision boundaries.

V. CONCLUSION

Addressing the long-tailed distribution and the continual emergence of new classes in real-world RSSC, this article investigates the problem of incremental classification for

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long-tailed remote sensing data and proposes CEF-CL, an incremental learning framework that continuously evolves both the feature backbone and classifier. The framework decouples the learning of the feature backbone and classifier, enhancing adaptability to long-tailed data through the integration of a tail-specific scaling grafting method and an adaptive multiscale head–tail feature fusion strategy in both the feature learning and classifier fine-tuning stages. In addition, by treating old classes as pseudo-tail classes during incremental training, CEF-CL effectively mitigates catastrophic forgetting.

In future work, we aim to enhance the rapid adaptability of CIL models to tail classes without degrading head-class performance, addressing challenges such as high intraclass variability in long-tailed remote sensing data. Furthermore, considering that new task data in real-world applications may originate from different geographical regions or temporal domains, leading to significant feature distribution shifts, maintaining the generalization performance of CIL models remains a critical research direction.

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